

## Coursework Cover Sheet

Students should complete the input fields contained in this form and attach it in front of your formal assessment submission. All fields within this form are required. Please ensure that check boxes and radio buttons are appropriately selected. The last three questions are just for you to personally consider.

### 1 Department and assessment information:

**School Name:** Engineering & Computing

**Assessment title:** Individual Report

**Course Title:** Bsc Software Engineering

**Module Title:** Computing Challenge

**Module Code:** CO1007

**Year of Study:** 2026

### 2 Academic Misconduct / Plagiarism Declaration

By attaching this front cover sheet to my assessment I confirm and declare that **I am the sole author of this work**, except where otherwise acknowledged by appropriate referencing and citation, and that I have taken all reasonable skill and care to ensure that no other person has been able, or allowed, to copy this work in either paper or electronic form, and that prior to submission I have read, understood and followed the University regulations as outlined in the [Academic Integrity Policy and Procedure for Academic Misconduct](#)

### 3 Have you checked the following? This will help your assessment achievement.

I have applied the learning outcomes for this module ☒

I have checked for Academic Integrity via Turn-it-in ☒

I have followed the guidance in the Assessment Brief and have not used AI to boost my grade unfairly. ☒

I have used references in accordance with instructions in the Assessment Brief ☒

I have proofread my work for spelling, grammar and punctuation. ☒

I have checked that the word count/size of this submissions accords with the guidance provided in the Assessment Brief. ☒

### 4 Well-being

We wish to support any student who is experiencing mitigating circumstances which prevents students from performing to the best of their ability when completing or submitting assignments. If you are experiencing such circumstances, then you may apply for Mitigating Circumstances. Wherever possible this must be done prior to handing in your assignment.

Do you need to apply for mitigating circumstances for this assignment ☐ No

Please refer to the [Mitigating Circumstances Policy](#)

#### 4.1 Questions you may wish to consider:

1. Have I allowed sufficient time to prepare this assessment? Yes
2. Have I reflected on previous feedback and made improvements in accordance with advice? Yes
3. What grade am I expecting? First Class



---

# CO1007 The Computing Challenge

---

Individual Portfolio



APRIL 27, 2026

STUDENT NUMBER: G21407227 (Tharushi Liyanage)

## Contents

|       |                                        |    |
|-------|----------------------------------------|----|
| 1     | Agile Methodology .....                | 3  |
| 1.1   | Agile Techniques .....                 | 3  |
| 1.2   | Stand-Up Meeting .....                 | 4  |
| 1.2.1 | Kanban Board .....                     | 4  |
| 1.2.2 | Stand-up meeting proof .....           | 5  |
| 1.2.3 | Stand-up meeting notes .....           | 5  |
| 2     | Design .....                           | 6  |
| 2.1   | App UI Design and UX .....             | 6  |
| 2.1.1 | Low-fidelity Wireframe .....           | 6  |
| 2.1.2 | High-fidelity Wireframe .....          | 7  |
| 2.2   | System Design .....                    | 8  |
| 3     | Implementation .....                   | 10 |
| 4     | Branding and Image .....               | 13 |
| 4.1   | Logo of the App .....                  | 13 |
| 4.2   | Poster of the app .....                | 14 |
| 4.3   | Viral Video .....                      | 15 |
| 4.4   | Social Media Marketing .....           | 15 |
| 5     | Testing .....                          | 16 |
| 5.1   | Test Log .....                         | 16 |
| 5.2   | UI/UX Usability Testing Feedback ..... | 18 |
| 6     | The Symposium .....                    | 19 |
| 6.1   | Our Team .....                         | 19 |
| 6.2   | Other teams .....                      | 19 |
| 7     | Reflection .....                       | 20 |
| 8     | References .....                       | 21 |

# 1 Agile Methodology

## 1.1 Agile Techniques

To build the Qodex Application we used the Agile software development methodology. This is based on building software in small parts called sprints and delivering it often, which is called Iterative and Incremental Development. This lets teams to get feedback early on and make changes as needed during the development process. Agile is different from traditional methods because it encourages the developers and stakeholders to work closely together. This makes sure that the final product meets the user requirements. Constant integration, frequent testing, and adaptive planning are some of the best ways that teams can keep their work in high quality while also being adaptive to changing quickly (Kumar & Bhatia, 2012). Because of this, agile is now the best way to work in environments that needs constant changing and moving quickly.

Another important part of Agile methods is how much they stress working together and talking to each other. Agile methods encourage team members to talk to each other in every stage of their development. Stand-up meetings, where progress and problems are talked about, are one way to do this. This constant communication among the stakeholders and the developers helps to find and fix problems quickly. Agile also promotes self-organization teams, where each member of the team is responsible for their own work and works with others to reach shared goals. This makes every member of the team more productive and makes sure that everyone is on the same page with the project's goals. Research indicated that fluent communication within agile teams markedly improves project success (Chan & Thong, 2008).

We chose the Kanban method to manage our projects. Kanban is a tool for managing the workflow to helps the teams to plan their work and find problems quickly and easily. The kanban board for Qodex project had four main columns: Backlog, To Do, In progress & and Completed. There was a card for each task, such as making user interfaces, adding quiz features, connecting to Google sheets, and testing the leaderboard. This visual structure made things clearer, made it easier to split up tasks, and cut down on work that was already being done.

The team also used an iterative approach, which meant they kept testing and making it system better after each step of development. Feedback was used to improve every part, from making the prototype to setting up server connections. This process helped find problems early on, improve the application performance and make it easier to use. In general, using both Agile and Kanban together made it easy, organized, and flexible to work on Qodex.

## 1.2 Stand-Up Meeting

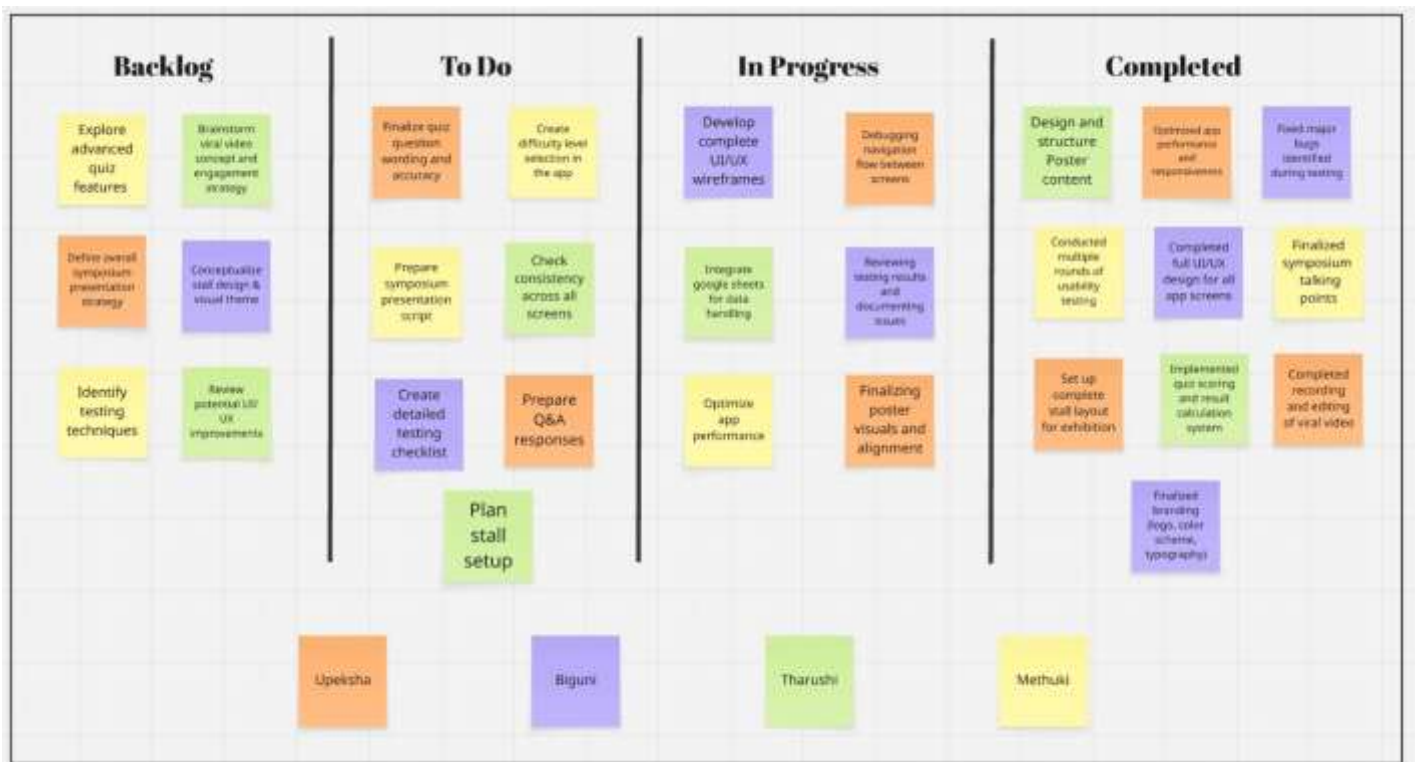
We had one of our team's stand-up meetings near the end of the development, when we were working on the leaderboard feature and improving the riddle system. The meeting was lasted for 10 to 15 minutes and went in a certain order. Everyone talked about what they had done so far, what they still needed to do, and any problems they were having. Our team leader ran the meeting and make sure that everyone had a chance to talk, and that the conversation stayed on track (Stray et al., 2016).

As a member of the team, I actively participated by giving updates on the tasks I was in charge of and bringing up a problem with the riddle-loading process being slow and the leaderboard issue. I was ready for the meeting because I had looked over my work ahead of time and found the main problem that was affecting with the leaderboard functionality. The conversation was useful and quick because everyone on the team was ready. Everyone at the meeting was able to talk to each other and work together clearly.

The group talked about the problem of loading riddles caused by the Session ID and came up with some possible fixes. We agreed to change the system logic to make sequential loading possible by talking about our ideas and looking at different options. This choice got rid of the delay and made the system work better in general. This shows that stand-up meetings help teams quickly find and fix problems by making sure everyone understands what there are responsible with (Stray et al., 2016).

People were more responsible and aware because of stand-up meetings. Members had to keep track of how others were doing while also keeping their own commitments. Such transparency kept things going and make sure that no one did the same task twice. The meeting was in line with Agile's core values and often led to good ideas for small changes.

### 1.2.1 Kanban Board



Kanban Board link: [https://miro.com/app/board/uXjVGnpBtVc=](https://miro.com/app/board/uXjVGnpBtVc=/)

## 1.2.2 Stand-up meeting proof



## 1.2.3 Stand-up meeting notes

| Team Standup     |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | 1ST DAY<br>15/03/2026                                                                                                                                                                                                                                                                                                                                                   | 2ND DAY<br>22/03/2026                                                                                                                                                                                                                                                                                 | 3 <sup>RD</sup> DAY<br>28/03/2026                                                                                                                                                                                                                                                          | 4TH DAY<br>03/04/2026                                                                                                                                                                                                                                                       | 5TH DAY<br>16/04/2026                                                                                                                                                                                                                                                          |
| <b>Biguni</b>    | <ul style="list-style-type: none"> <li>Started designing the main screens of the app or Thinkable, including the welcome page, quiz page, and result page.</li> <li>Arranged the navigation between screens so the app flow is easier to understand.</li> <li>Made some small UI changes like adjusting button placements and choosing a simple color theme.</li> </ul> | <ul style="list-style-type: none"> <li>Improved the design of the quiz screen by arranging labels and answer buttons more clearly.</li> <li>Updated text sizes and spacing so the questions are easier to read.</li> <li>Continued making the app interface look more neat and consistent.</li> </ul> | <ul style="list-style-type: none"> <li>Made a few improvements to the app design based on early feedback.</li> <li>Adjusted some colors and text visibility to make the interface clearer.</li> <li>Updated the result screen so the final score can be displayed more clearly.</li> </ul> | <ul style="list-style-type: none"> <li>Finalized most of the app design and made the screens look more consistent.</li> <li>Improved spacing, alignment, and overall appearance of the app.</li> <li>Added small finishing touches to make the app look cleaner.</li> </ul> | <ul style="list-style-type: none"> <li>Reviewed the full app design one last time and made a few small corrections.</li> <li>Checked if all screens looked ready for the final demonstration.</li> <li>Made sure the app had a clean and simple appearance overall.</li> </ul> |
| <b>Methuki</b>   | <ul style="list-style-type: none"> <li>Began working on the quiz logic using Thinkable blocks.</li> <li>Connected the start button to move from the home screen to the quiz screen.</li> <li>Planned how the quiz questions will appear one by one.</li> </ul>                                                                                                          | <ul style="list-style-type: none"> <li>Worked on the next button logic to move between quiz questions.</li> <li>Connected the quiz screen to the result screen after the last question.</li> <li>Started adding blocks to recognize which answer the user selects.</li> </ul>                         | <ul style="list-style-type: none"> <li>Continued working on the answer-checking logic.</li> <li>Connected selected answers with the score system.</li> <li>Fixed a few issues with moving to the next question properly.</li> </ul>                                                        | <ul style="list-style-type: none"> <li>Fixed bugs related to navigation and answer selection.</li> <li>Improved the smoothness of the app when moving between questions and screens.</li> <li>Double-checked that all buttons are working as expected.</li> </ul>           | <ul style="list-style-type: none"> <li>Tested the app logic one final time.</li> <li>Verified that navigation between all screens worked properly.</li> <li>Confirmed that there were no major issues in the quiz flow.</li> </ul>                                             |
| <b>Tharushi</b>  | <ul style="list-style-type: none"> <li>Prepared the quiz questions and answer options for the app.</li> <li>Sorted the questions into categories to make them easier to organize.</li> <li>Started noting down the correct answers for scoring later.</li> </ul>                                                                                                        | <ul style="list-style-type: none"> <li>Added the prepared quiz questions and answer choices into Thinkable.</li> <li>Started creating the score logic so the app can count correct answers.</li> <li>Helped organize the order of questions for a smoother quiz flow.</li> </ul>                      | <ul style="list-style-type: none"> <li>Completed most of the score calculation logic.</li> <li>Added a restart option so users can try the quiz again after finishing.</li> <li>Checked if the score resets correctly when the quiz restarts.</li> </ul>                                   | <ul style="list-style-type: none"> <li>Reviewed the quiz questions and score tracking logic again.</li> <li>Made sure all questions load correctly without errors.</li> <li>Cleaned up some extra or unnecessary blocks in the app.</li> </ul>                              | <ul style="list-style-type: none"> <li>Rechecked all quiz questions, answers, and score calculations.</li> <li>Tested the restart and result screen functions again.</li> <li>Confirmed that everything was ready for final submission.</li> </ul>                             |
| <b>Upeksha</b>   | <ul style="list-style-type: none"> <li>Reviewed the project requirements and listed the main features we need to complete.</li> <li>Created a simple checklist for testing the app later.</li> <li>Looked at a few quiz app examples to get ideas for the layout and features.</li> </ul>                                                                               | <ul style="list-style-type: none"> <li>Tested the screen changes and button navigation.</li> <li>Noticed a few small issues with alignment and screen transitions.</li> <li>Took screenshots of the app for future testing evidence and report use.</li> </ul>                                        | <ul style="list-style-type: none"> <li>Carried out the first round of testing with a few classmates/friends.</li> <li>Collected feedback about the design, app flow, and button behavior.</li> <li>Started updating the project notes with testing results and screenshots.</li> </ul>     | <ul style="list-style-type: none"> <li>Tested the app several times from start to finish.</li> <li>Checked whether the final score displays correctly after completing the quiz.</li> <li>Organized the testing notes and prepared them for the report.</li> </ul>          | <ul style="list-style-type: none"> <li>Finalized the testing evidence and screenshots.</li> <li>Helped complete the report and review all project sections.</li> <li>Assisted with preparing the presentation and app explanation.</li> </ul>                                  |
| <b>Next step</b> | <ul style="list-style-type: none"> <li>Finalize the full set of quiz questions.</li> <li>Continue working on navigation between screens.</li> <li>Start building the answer selection process.</li> </ul>                                                                                                                                                               | <ul style="list-style-type: none"> <li>Complete the scoring system.</li> <li>Add answer feedback if possible.</li> <li>Start testing the quiz from beginning to end.</li> </ul>                                                                                                                       | <ul style="list-style-type: none"> <li>Fix the issues found during testing.</li> <li>Improve the app flow and user experience.</li> <li>Get ready for final testing.</li> </ul>                                                                                                            | <ul style="list-style-type: none"> <li>Finalize the report and screenshots.</li> <li>Run one complete app demo before submission.</li> <li>Prepare for the project presentation.</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>Submit the final report and presentation.</li> <li>Demonstrate the Thinkable Quiz App successfully.</li> <li>Keep all project materials ready for review if needed.</li> </ul>                                                          |

## 2 Design

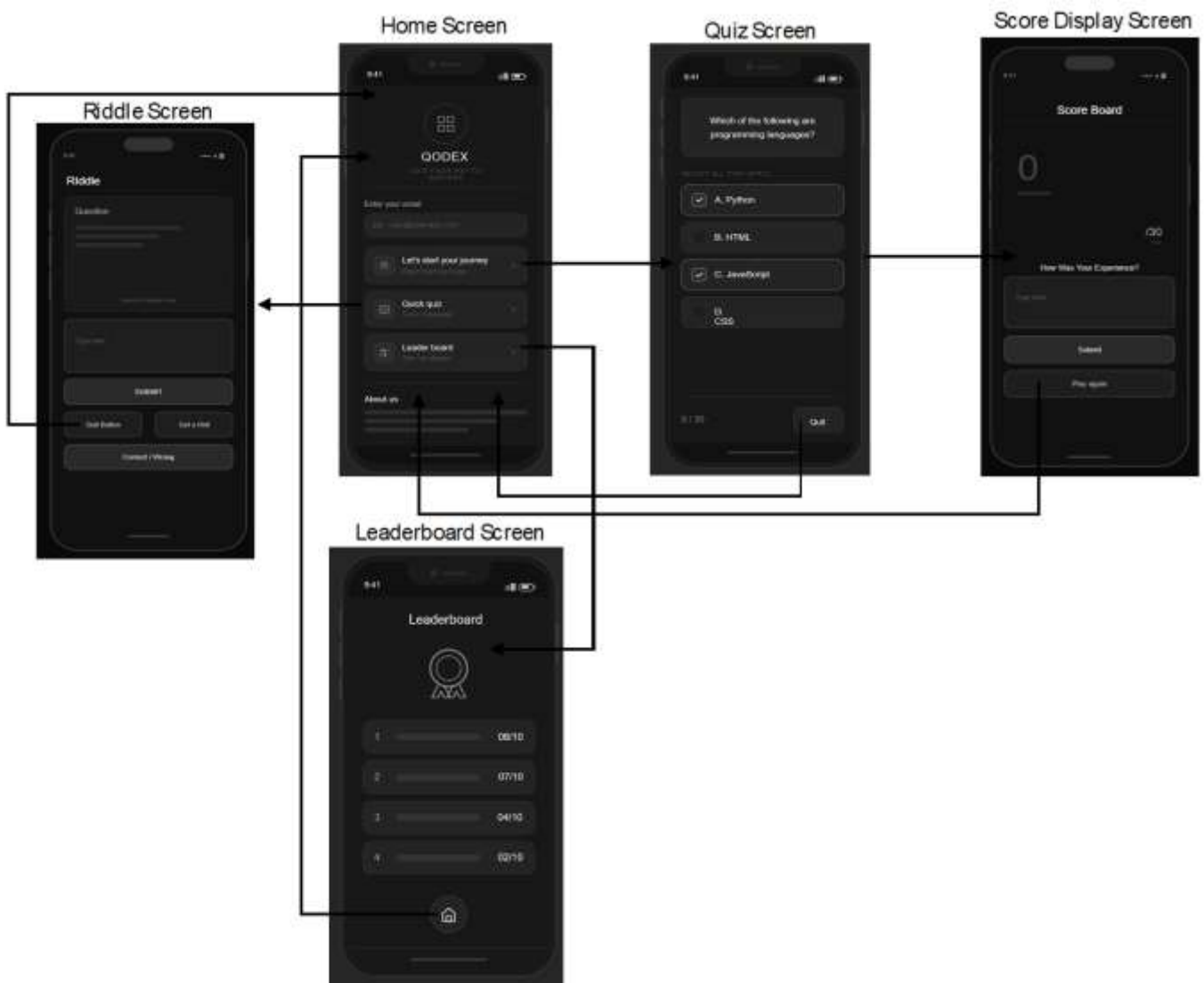
### 2.1 App UI Design and UX

The Qodex app was designed through a process of trial that used both the low-fidelity wireframes and high-fidelity UI mock-ups. The wireframes helped organise the structure and flow of the navigation between the important screens such as the home screen, riddle screen, quiz screen, leaderboard, and score display.

The final UI, on the other hand, gave the app a consistent look with a purple theme and easy-to-read text. The interface design follows well-known UI rules, such as making things clear about the system's status, consistent, and easy to use. According to the Chen et al., 2020; good interfaces should make things easier for users and show information in a way that is clear and easy to understand. Qodex makes it easy for users to switch between tasks by clearly grouping actions (like starting a quiz or looking at the leaderboard) and make it easy to secure.

1. Home screen → Enter the email & Access to the “Lets Start your journey”, “Quick Quiz”, “Leaderboard”
2. Quick Quiz → General Knowledge multiple-choice questions from Google Sheets
3. Lets Start Your Journey → Riddles fetched from remote server using session IDs
4. Score Display → After the MCQ questions it displays your marks out of 30
5. Leaderboard → Top scores for competition

#### 2.1.1 Low-fidelity Wireframe



### 2.1.2 High-fidelity Wireframe



According to the Natarajan, Janani (2025); having same layouts, terms, and ways of interacting throughout an app's interface helps users to build reliable, meaningful models. This makes it easier to find their way around and requires less Cognitive effort.

Key Components,

- ❖ App was built-in user-friendly language like “Start”, “Get a hint”, “Submit” and etc.
- ❖ We have given user control freedom across all the screens. Ex: “Quit”, “Exit”.
- ❖ It has the consistency in the background colour, theme, and button colours.
- ❖ All options are visible properly.

By these components, it ensures that the app was structured and functional for enjoyable learning experiences.

## 2.2 System Design

The way Qodex is designed makes it clear that the front end and back end are separate. This fits with the client-server architecture. The front end is the mobile app that was made with Thunkable. The wireframe screens show the home screen, the screen with the riddle, the screen with the quiz, the screen with the score and the screen with the leaderboard. This layer is in charge of user interaction by showing the MCQ questions, getting riddles, and showing results. Features like “Let’s start your journey” and “Quick quiz” make it easy and intuitive for users to get around the site. These features make it easy to navigate and interact with the app. The front-end refers to the client-side interface in system design that makes complex processes easier in a visual experience that is user friendly and allows for smooth navigation and usability (Boland, 1978).

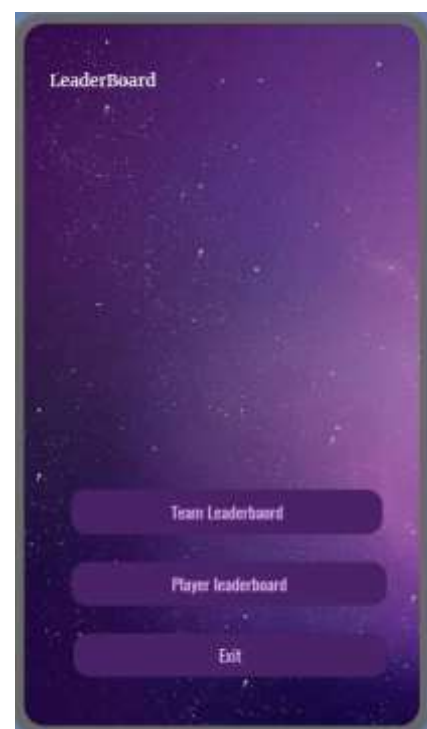
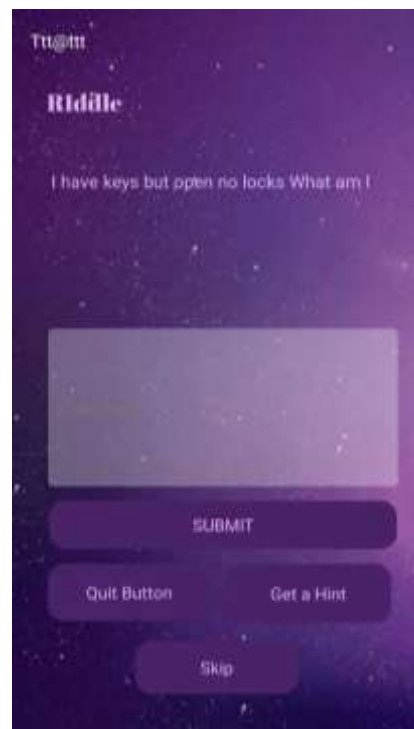
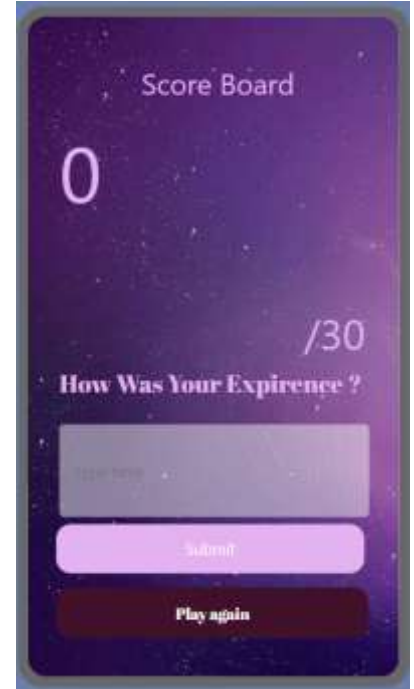
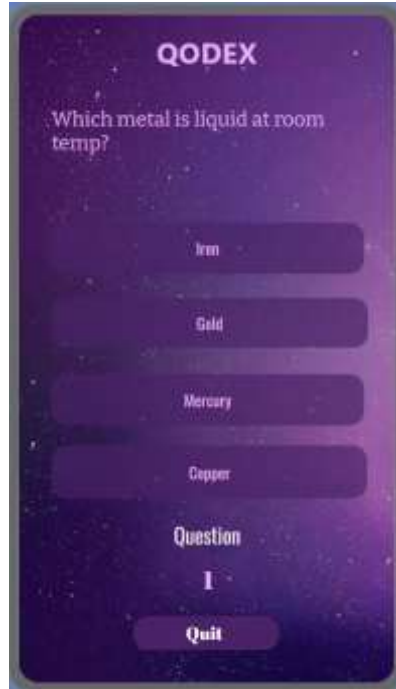
The back end works behind the scenes to manage the data storage, processing and system logic. In Qodex this means getting MCQ questions from the Google sheet and processing user answers through the Web API. The back end is the part of an application that users don’t see but ensures that the application works properly by handling databases, server communication and application logic. When the user clicks a button the front-end send a request to the back end and the back end responds. This is a good design as it separates the data and business logic from the presentation layer which can improve the maintainability and performance of the system.

State transition diagram should show:

- ❖ Initial state → Home screen
- ❖ Home → “Let’s start your Journey”, “Quick Quiz” Mode selection
- ❖ Let’s start your Journey → Question loading → Select the correct answer → Score Calculation → Results
- ❖ Show results screen → Replay Option
- ❖ Quick Quiz → Riddle request (with the session ID) → Display the riddle Question → Get a hint or submit the answer → Verify the answer with the score → Next riddle → Results
- ❖ Leaderboard → Player Leaderboard → Show the results from the server.
- ❖ Leaderboard → Team Leaderboard

### Questions From the Google Sheet

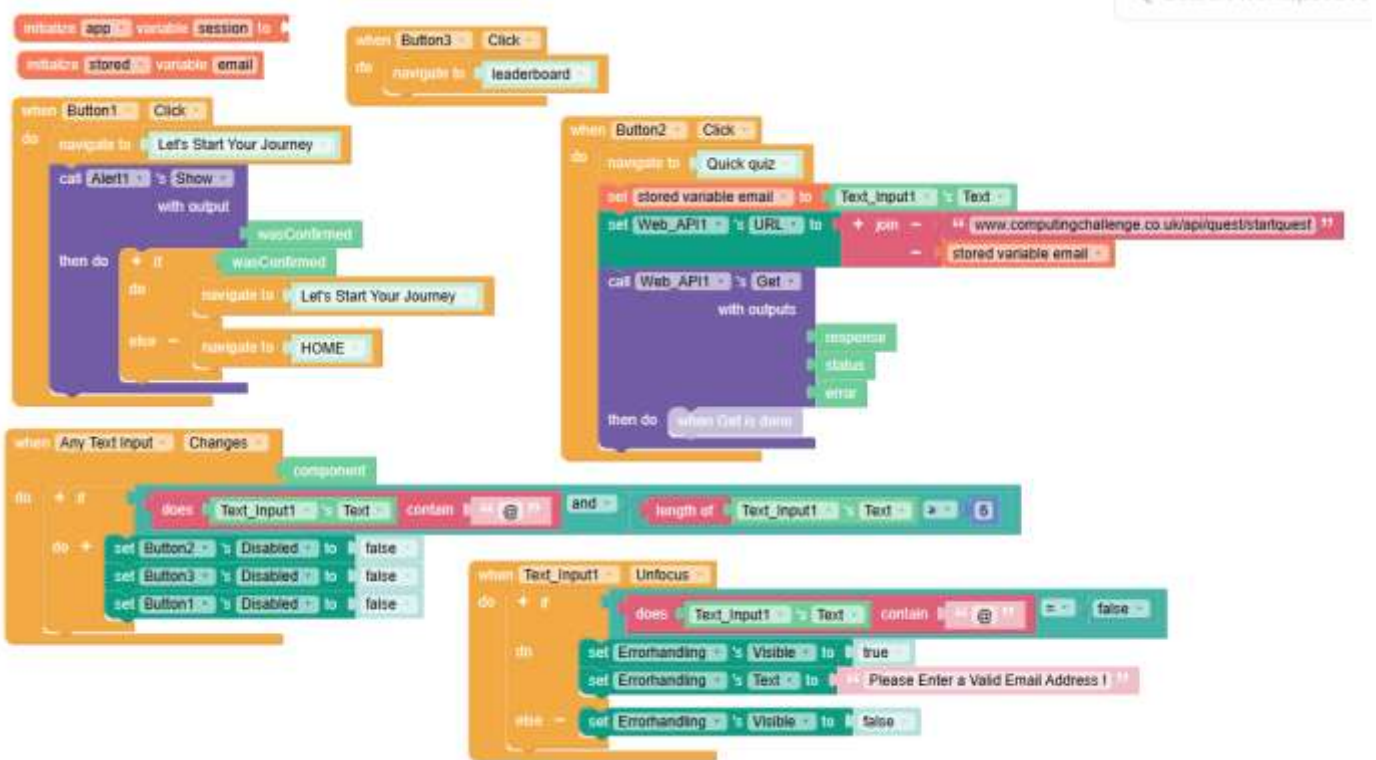
|    | Unnamed Column 1                                 | 1           | 2         | 3              | 4             | Correct Answer |
|----|--------------------------------------------------|-------------|-----------|----------------|---------------|----------------|
| 1  | What is the capital of Japan?                    | Tokyo       | Beijing   | Seoul          | Bangkok       | 1              |
| 2  | Which planet is the largest?                     | Earth       | Mars      | Jupiter        | Venus         | 3              |
| 3  | Who wrote Hamlet?                                | Shakespeare | Twain     | Austen         | Dickens       | 1              |
| 4  | What is the longest river in the world?          | Nile        | Amazon    | Yangtze        | Mississippi   | 1              |
| 5  | Which country has the Great Wall?                | India       | China     | Japan          | Korea         | 2              |
| 6  | What is the hardest natural substance?           | Gold        | Iron      | Diamond        | Silver        | 3              |
| 7  | Who invented the telephone?                      | Edison      | Bell      | Tesla          | Newton        | 2              |
| 8  | Which continent is Brazil in?                    | Asia        | Europe    | Africa         | South America | 4              |
| 9  | What is the capital of Canada?                   | Toronto     | Ottawa    | Vancouver      | Montreal      | 2              |
| 10 | Which animal is known as the King of the Jungle? | Tiger       | Lion      | Elephant       | Leopard       | 2              |
| 11 | Which gas do plants absorb?                      | Oxygen      | Nitrogen  | Carbon Dioxide | Hydrogen      | 3              |
| 12 | What is the smallest continent?                  | Asia        | Australia | Europe         | Antarctica    | 2              |
| 13 | Who painted the Mona Lisa?                       | Van Gogh    | Picasso   | Da Vinci       | Rembrandt     | 3              |
| 14 | Which ocean is the largest?                      | Atlantic    | Indian    | Pacific        | Arctic        | 3              |
| 15 | What is the capital of Italy?                    | Rome        | Milan     | Naples         | Venice        | 1              |
| 16 | Which planet is closest to the sun?              | Earth       | Venus     | Mercury        | Mars          | 3              |
| 17 | What is the currency of the UK?                  | Dollar      | Euro      | Pound          | Yen           | 3              |
| 18 | Who discovered America?                          | Columbus    | Magellan  | Cook           | Vespucci      | 1              |
| 19 | What is the tallest mountain?                    | K2          | Everest   | Alps           | Kilimanjaro   | 2              |
| 20 | Which country is famous for pyramids?            | India       | Mexico    | Egypt          | Peru          | 3              |
| 21 | What is the capital of Spain?                    | Barcelona   | Madrid    | Seville        | Valencia      | 2              |
| 22 | Which bird cannot fly?                           | Eagle       | Penguin   | Sparrow        | Parrot        | 2              |
| 23 | What is the fastest land animal?                 | Lion        | Tiger     | Cheetah        | Leopard       | 3              |
| 24 | Which organ pumps blood?                         | Brain       | Heart     | Liver          | Lungs         | 2              |
| 25 | Which country is known for sushi?                | China       | Japan     | Thailand       | Vietnam       | 2              |
| 26 | What is the capital of Germany?                  | Berlin      | Munich    | Frankfurt      | Hamburg       | 1              |
| 27 | Which metal is liquid at room temp?              | Iron        | Gold      | Mercury        | Copper        | 3              |
| 28 | Which desert is the largest?                     | Sahara      | Gobi      | Kalahari       | Arabian       | 1              |
| 29 | What is the capital of Australia?                | Sydney      | Melbourne | Canberra       | Perth         | 3              |
| 30 | Which sport uses a racket?                       | Football    | Tennis    | Cricket        | Hockey        | 2              |



### 3 Implementation

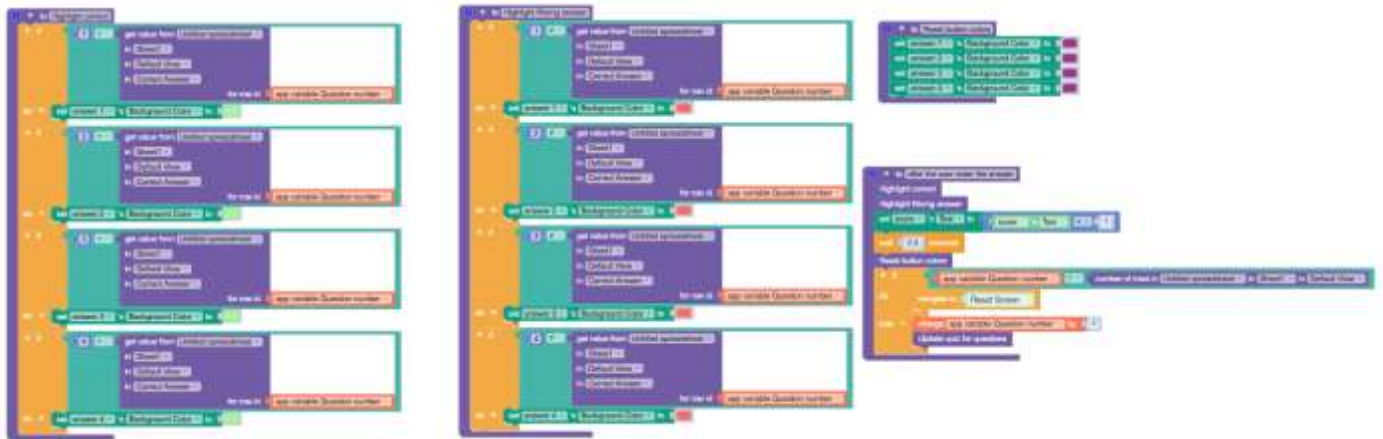
The team app is composed of user interface elements and structures block code that enables smooth client-server communication. The flow of the quiz is made using buttons, labels, text inputs, Web API calls. The user enters the email and clicks start, the app checks the input, constructs a server URL, and starts the quest through a Web API block and saves the session ID for tracking. The server replies with a JSON object containing a riddle. This is parsed by “get property of object” blocks and then displayed on a label. When the user submits an answer, another Web API calls sends the solution to the server and the score variable is updated. If-then control blocks make sure that the validation is done correctly. Navigation blocks make it easier for the user to get around the quiz, results, and leaderboard screens. Each sequence is made up of modular blocks that make it easy to see who is in charge of input, server communication, and user feedback. This makes the app interactive.

**Screen 1: Home Screen**



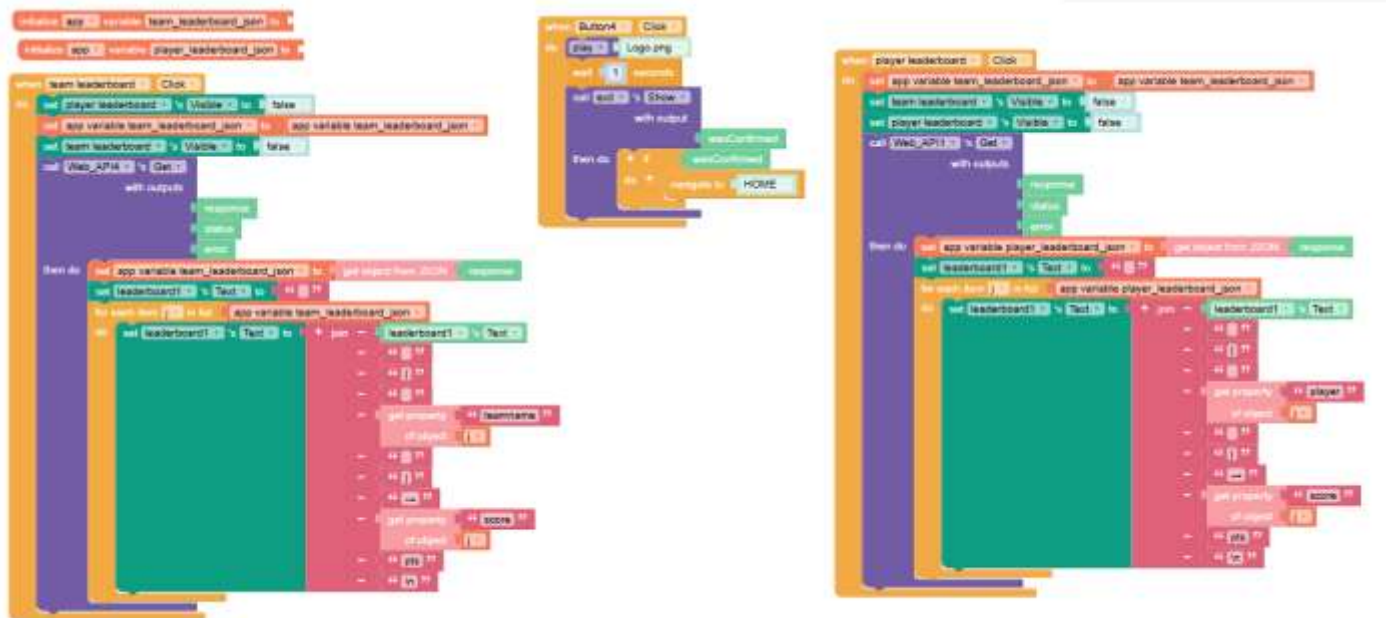
**Screen 2: Let's Start Your Journey**





Screen 3: Quick Quiz





## 4 Branding and Image

The idea behind QODEX branding were innovation, intelligence and learning that was fun. The logo shows a robot mascot from the future inside the letters. It represents technology, knowledge, and the idea of digital quizzes as a whole. The team name QODEX is a blend of "code" and "index", suggesting a structured, tech-savvy quiz platform. The stylised lettering reinforces the modern and digital identity of the brand. The logo's modern circular shape makes it look both professional and fun, which makes the app easy to spot as well as excellent for kids.

The purple colours were meant to show off our brand's creativity, wisdom, and originality, while the neon touches made the app even more bright and modern. We wanted to make a visual style that would match the interactive quiz game Qodex in this way. We also used a catchy slogan "Quiz Your Way to Success", to help the audience remember our brand and emphasized the educational value of our app.

To make our brand look better, we made a video to promote it. It has animated transitions, and footage of gameplay. This kind of branding helped us stand out.

### 4.1 Logo of the App



## 4.2 Poster of the app

**QODEX**  
QUIZ YOUR WAY TO SUCCESS

✦ Riddles & Quizzes · Points · Leaderboard

**What is Qodex?**  
Qodex is a riddle & quiz-based mobile app designed to make learning fun, interactive, and engaging.  
Users solve riddles, answer quiz questions, earn scores, and compete on a leaderboard.

**Why was it created?**  
Qodex was built to transform passive studying into active, gamified challenges — keeping learners motivated and making every session feel like a game rather than a chore.

**How does it work?**

Home Riddle Quiz Score Leaderboard

Enter your email to start Choose Riddle or Quick Quiz Answer questions Use a hint if stuck  
Quit anytime View your score Leave feedback Play again Check the leaderboard

**Key Benefits**

- ✓ Improves knowledge & retention
- ✓ Develops critical thinking
- ✓ Enhances problem-solving
- ✓ Keeps learners motivated
- ✓ Instant score feedback every round

**What makes it special?**

- ✦ Two modes: Riddles & Quick Quizzes
- ✦ Hint system to guide learners
- ✦ Real-time score tracking
- ✦ Global leaderboard ranking
- ✦ Post-quiz experience feedback

**Main Purpose**  
Deliver an engaging learning experience through riddles and quizzes — helping users test knowledge, sharpen thinking, and track progress via scores and rankings.

**Continuous Learning**  
Users can replay quizzes and riddles anytime to improve their score. The leaderboard drives healthy competition and encourages repeated practice.

**Skill Building**  
Riddles sharpen logical reasoning and creative thinking. Quizzes reinforce recall and active learning — building a sharper, more confident mind.

**Motivation & Engagement**

Compete on the leaderboard Track your score out of 30 Use hints to stay in the game  
Correct / wrong answer feedback Share your experience after each round Play again to beat your score

**Our Team**

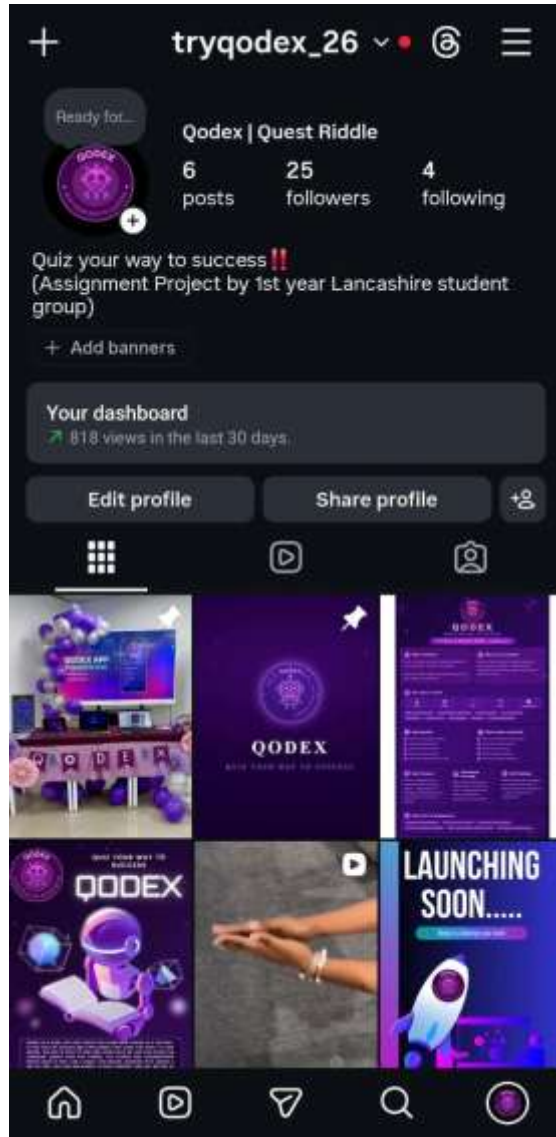
University of Lancashire UCL

Upeksha Biguni Tharushi Methuki

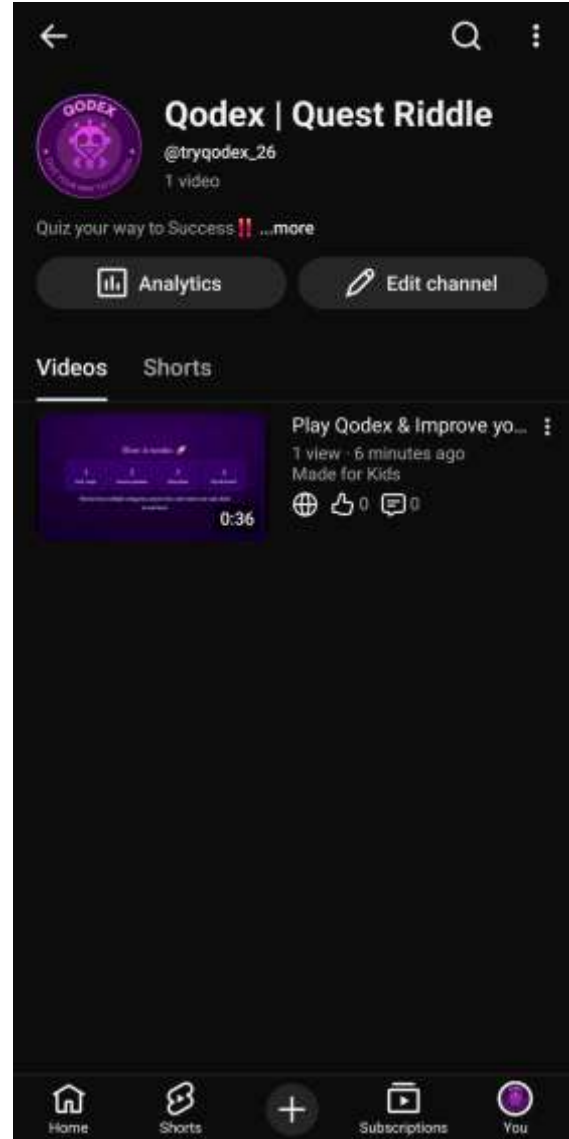
### 4.3 Viral Video

[https://www.youtube.com/watch?v=dpXhJ\\_FU-cl](https://www.youtube.com/watch?v=dpXhJ_FU-cl)

### 4.4 Social Media Marketing



Instagram



YouTube

## 5 Testing

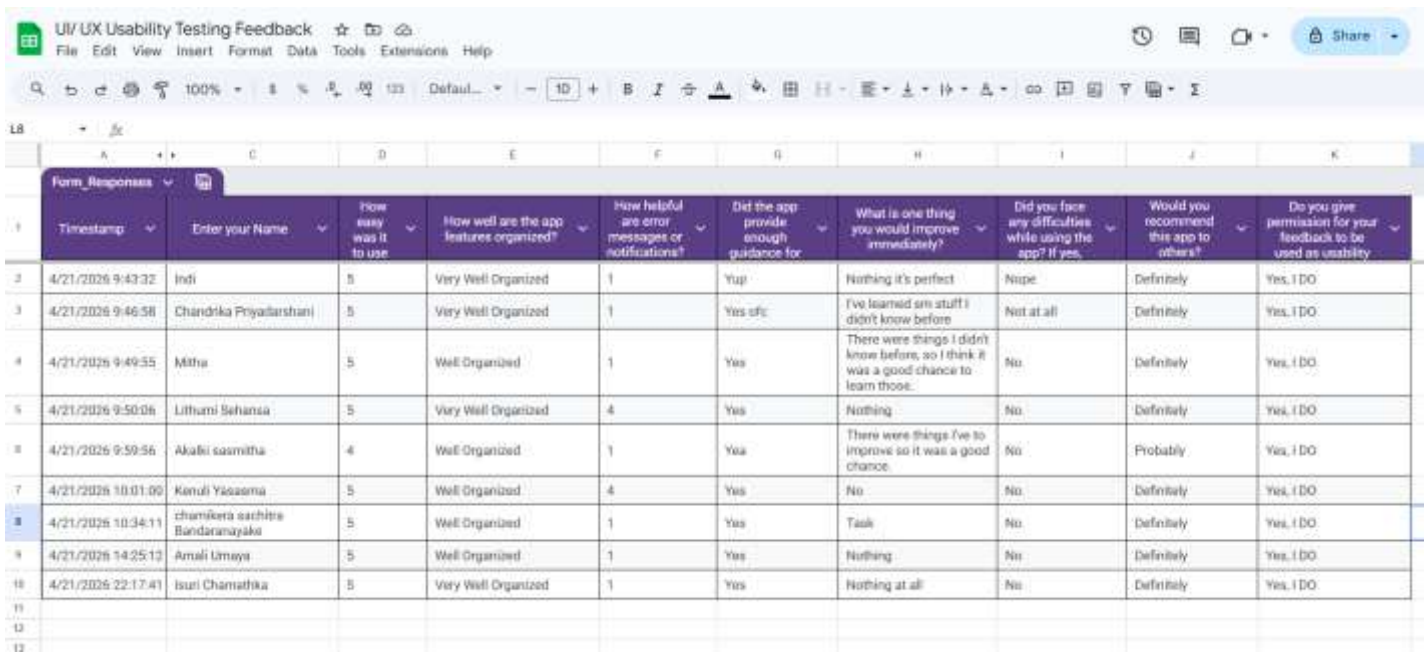
Testing was done in a structured, step-by-step way to make sure that each stage of development was in high quality by following an iterative methodology. We did functional testing, usability testing, and compatibility checks on different devices and in different setting. Testing on android devices and desktop preview setups showed that the app worked well, and the user interface worked the same way every time. It was confirmed that core features like loading quiz data, getting riddles from the server, and updating leaderboard scores worked reliably. User testing sessions with classmates and peers gave us useful information about how clear the interface was, how easy it was to navigate, and how usable the app was overall. This helped us improve the app experience. This method of constant testing made sure that both technical accuracy and user satisfaction were met.

### 5.1 Test Log

| Test Case                            | Feature               | Test Input/ Action                    | Expected Results                                                        | Actual Result | Pass/ Fail |
|--------------------------------------|-----------------------|---------------------------------------|-------------------------------------------------------------------------|---------------|------------|
| <b>App Interface</b>                 |                       |                                       |                                                                         |               |            |
| 01.                                  | App Launch            | Open app on device                    | Home screen displays the buttons properly                               | As expected   | Pass       |
| 02.                                  | Animation of the app  | Open app on device                    | Show the animations on the app                                          | As expected   | Pass       |
| 03.                                  | Page Scrolling        | Scrolling                             | Scroll the App interface                                                | As expected   | Pass       |
| 04.                                  | Start the Application | Scroll the page                       | Scroll the page (Lets start your journey here, Quick Quiz, Leaderboard) | As expected   | Pass       |
| 05.                                  | User Email            | Enter the email                       | User only can play the game after input a valid email address           | As expected   | Pass       |
| <b>Let's Start Your Journey Here</b> |                       |                                       |                                                                         |               |            |
| 06.                                  | Showing the question  | Press "Let's start your journey here" | Show the question and the answer                                        | As expected   | Pass       |
| 07.                                  | Answer Selection      | Select and answer option              | Selected answer is highlighted and recorded                             | As expected   | pass       |
| 08.                                  | Score calculation     | Complete the quiz                     | Final score is calculated and displayed correctly                       | As expected   | Pass       |
| <b>Quick Quiz</b>                    |                       |                                       |                                                                         |               |            |
| 09.                                  | Quick Quiz Launch     | Tap "Quick Quiz" on Home Screen       | Quiz starts in challenge mode                                           | As expected   | Pass       |
| 10.                                  | Question Display      | Observe quiz question                 | Question text shown clearly with 4 options                              | As expected   | Pass       |
| 11.                                  | Answer and Next       | Select answer and proceed             | Move to next question after answer selected                             | As expected   | Pass       |
| 12.                                  | Score on Completion   | Complete all questions                | Final score shown on Score Display Screen                               | As expected   | Pass       |
| 13.                                  | Quit During Quiz      | Tap "Quit" button mid-quiz            | User returns to Home Screen                                             | As expected   | Pass       |
| <b>Leaderboard</b>                   |                       |                                       |                                                                         |               |            |
| 14.                                  | Getting leaderboard   | Open leaderboard section              | Display top scores of users                                             | As expected   | Pass       |
| 15.                                  | Home navigation       | Tap Home button                       | Returns to home screen                                                  | As expected   | Pass       |

| <b>Riddle screen</b>        |                         |                              |                                                        |             |      |
|-----------------------------|-------------------------|------------------------------|--------------------------------------------------------|-------------|------|
| 16.                         | Riddle Question Display | Navigation to Riddle screen  | Question shown in display area                         | As expected | Pass |
| 17.                         | Answer Input            | Type answer in input field   | Input field accepts typed text answer                  | As expected | Pass |
| 18.                         | Submit Answer           | Tap submit with valid answer | Correct/Wrong feedback displayed                       | As expected | Pass |
| 19.                         | Empty submit Validation | Tap submit with blank input  | Error shown: User couldn't able to go to the next step | As expected | Pass |
| 20.                         | Get a hint              | Tap "Got a Hint" button      | Hint text revealed to assist user                      | As expected | Pass |
| 21.                         | Quit Riddle             | Tap Quit button              | Returns user to home screen                            | As expected | Pass |
| <b>Score Display Screen</b> |                         |                              |                                                        |             |      |
| 22.                         | Score Display           | Complete quiz and view score | Current score shown prominently on screen              | As expected | Pass |
| 23.                         | Score Format            | Observe score on screen      | Score displayed as X/30 format                         | As expected | Pass |
| 24.                         | Feedback Input          | Type in experience           | User can type feedback text                            | As expected | Pass |
| 25.                         | Feedback input          | Type in experience           | User can type feedback text                            | As expected | Pass |
| 26.                         | Submit Feedback         | Tap submit button            | Feedback submitted and confirmation shown              | As expected | Pass |
| 27.                         | Play again              | Tap "Play again" button      | Returns user to Quiz screen for new screen             | As Expected | Pass |

## 5.2 UI/UX Usability Testing Feedback



The screenshot shows a Google Sheet titled "UI/UX Usability Testing Feedback". The sheet contains a table with 10 columns: Timestamp, Enter your Name, How easy was it to use, How well are the app features organized?, How helpful are error messages or notifications?, Did the app provide enough guidance for, What is one thing you would improve immediately?, Did you face any difficulties while using the app? If yes, Would you recommend this app to others?, and Do you give permission for your feedback to be used as usability. The table lists 10 responses from users, with ratings for ease of use and organization, and qualitative feedback on helpfulness, guidance, and recommendations.

|    | Timestamp          | Enter your Name                 | How easy was it to use | How well are the app features organized? | How helpful are error messages or notifications? | Did the app provide enough guidance for | What is one thing you would improve immediately?                                        | Did you face any difficulties while using the app? If yes, | Would you recommend this app to others? | Do you give permission for your feedback to be used as usability |
|----|--------------------|---------------------------------|------------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------|
| 2  | 4/21/2026 9:43:32  | Indi                            | 5                      | Very Well Organized                      | 1                                                | Yup                                     | Nothing it's perfect                                                                    | Nope                                                       | Definitely                              | Yes, I DO                                                        |
| 3  | 4/21/2026 9:46:58  | Chandrika Priyadarshani         | 5                      | Very Well Organized                      | 1                                                | Yes sfc                                 | I've learned sm stuff I didn't know before                                              | Not at all                                                 | Definitely                              | Yes, I DO                                                        |
| 4  | 4/21/2026 9:49:55  | Mitha                           | 5                      | Well Organized                           | 1                                                | Yes                                     | There were things I didn't know before, so I think it was a good chance to learn those. | No                                                         | Definitely                              | Yes, I DO                                                        |
| 5  | 4/21/2026 9:50:06  | Lithani Sahansa                 | 5                      | Very Well Organized                      | 4                                                | Yes                                     | Nothing                                                                                 | No                                                         | Definitely                              | Yes, I DO                                                        |
| 6  | 4/21/2026 9:59:56  | Akshita sasmitha                | 4                      | Well Organized                           | 1                                                | Yes                                     | There were things I've to improve so it was a good chance                               | No                                                         | Probably                                | Yes, I DO                                                        |
| 7  | 4/21/2026 10:01:09 | Kendri Yasaama                  | 5                      | Well Organized                           | 4                                                | Yes                                     | No                                                                                      | No                                                         | Definitely                              | Yes, I DO                                                        |
| 8  | 4/21/2026 10:34:11 | chamikera sashita Bandaranayake | 5                      | Well Organized                           | 1                                                | Yes                                     | Task                                                                                    | No                                                         | Definitely                              | Yes, I DO                                                        |
| 9  | 4/21/2026 14:25:13 | Amali Umaya                     | 5                      | Well Organized                           | 1                                                | Yes                                     | Nothing                                                                                 | No                                                         | Definitely                              | Yes, I DO                                                        |
| 10 | 4/21/2026 22:17:41 | Isuri Chamathika                | 5                      | Very Well Organized                      | 1                                                | Yes                                     | Nothing at all                                                                          | No                                                         | Definitely                              | Yes, I DO                                                        |

We used a Google Form to get the feedback from the external people who used our application and ask them to comment on it using this link.

Google Form: [https://docs.google.com/forms/d/1pM9n2kT\\_F3fs2t3xukiJuWvV8ln6euLW6qOiZSYMJk4/edit](https://docs.google.com/forms/d/1pM9n2kT_F3fs2t3xukiJuWvV8ln6euLW6qOiZSYMJk4/edit)

Overall, the testing phase confirmed that Qodex was user-friendly, effective, and consistent across all the screens. Future improvements will be include adding the faster data retrieval and sound effects to enhance the user engagement.

## 6 The Symposium

Our team had a great time and learned a lot at the symposium. We set up our stall to show off the QODEX app, and we made sure it was eye-catching by putting up balloons, rosettes, and a bright banner. The app's interface was shown on the laptops and tablet, and the big monitor let us talk more about the app's features. As presenters, we worked together to show how to use the app, what it was for, and how it was made. We made sure that people could use the demo and ask questions. One of our strengths was that we worked well together. As a group from the day one to until the end of the symposium we all worked so hard. Everyone had a clear idea of what they had to do in the symposium, like explaining the technical parts, talking to the audience, or taking care of the visuals. We would make our stall more interactive, perhaps a live demo or a short video walkthrough, so visitors would stay longer, and we could improve next time.

There were many amazing and creative stalls from our peers. One team that caught my attention was one that showcased an app called Ezuya. The app has a simple, sleek design and practical features. Their stall was simple but effective, featuring clear posters and a hands-on demo to let visitors try out the app themselves. Overall, the symposium was a great learning experience.

### 6.1 Our Team



### 6.2 Other teams



## 7 Reflection

As a student in the computing challenge, I can reflect on my experience using Gibbs' Reflective Cycle. The stage of description is where I liked the most, presenting our project at the symposium and interactive with peers. The atmosphere was alive, and I enjoyed the opportunity to demonstrate our creatively decorated stall backed up with digital demonstrations. We felt a sense of confidence and pride in the work we were doing. But on the other hand, I had problems in handling the time especially judging preparation of the stall with other academic commitments. According to Gibbs (1988), reflection has to acknowledge the positive and negative, for it to be meaningful.

Moving to the feelings stage, I was excited and thrilled during the event. My primary duty on the team was to explain the Qodex presentation and ensure that our stand was friendly, in addition to assisting teammate with any questions. Positive aspects of our team included strong teamwork and effective communication, while negative aspects included occasional disagreements over design choices. Gibbs (1988) asserts that reflecting on emotions facilitates understanding how they influenced conduct.

By having open discussion about problems and signing tasks according to strengths, our team overcomes obstacles during the evaluation process. We were able to work effectively together toward a common goal. We naturally included elements of cooperative learning, where each participant offered a unique contribution, even though we didn't follow a formal teamwork model. For instance, even though our booth had a pleasing appearance, we might have increased interaction by adding more practical examples.

In the end, I realized that effective teamwork requires both structure and flexibility during the analysis and action plan stages. Overall, I was able to critically analyse my involvement using Gibb's Reflective cycle, indicating my strengths, areas for improvement, and challenges for upcoming group initiatives (Gibbs, 1988).

## 8 References

- Baresi, L., Di Nitto, E. and Ghezzi, C. (2008) ‘Toward open-world software: Issue of software engineering’, *Decision Support Systems*, 45(2), pp. 375–386. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0167923608002133> (Accessed: 10 April 2026).
- Kumar, G. (2013) ‘Impact of Agile Methodology on Software Development Process’, *ResearchGate*. Available at: [https://www.researchgate.net/profile/Gaurav-Kumar-175/publication/255707851\\_Impact\\_of\\_Agile\\_Methodology\\_on\\_Software\\_Development\\_Process/links/00b49520489442e12d000000/Impact-of-Agile-Methodology-on-Software-Development-Process.pdf](https://www.researchgate.net/profile/Gaurav-Kumar-175/publication/255707851_Impact_of_Agile_Methodology_on_Software_Development_Process/links/00b49520489442e12d000000/Impact-of-Agile-Methodology-on-Software-Development-Process.pdf) (Accessed: 16 April 2026).
- Petersen, K., Wohlin, C. and Baca, D. (2016) ‘The waterfall model in large-scale development’, *Journal of Systems and Software*, 122, pp. 1–16. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0164121216000066> (Accessed: 16 April 2026).
- Ralph, P. and Kelly, P. (2020) ‘The dimensions of software engineering’, *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW2), pp. 1–23. Available at: <https://dl.acm.org/doi/abs/10.1145/3391613> (Accessed: 24 April 2026).
- Hämmäläinen, J. (2021) *Agile Transformation in Software Development Companies*. Doctoral dissertation, University of Eastern Finland. Available at: <https://www.doria.fi/handle/10024/192677> (Accessed: 16 April 2026).
- Tushman, M.L. and Nadler, D.A. (1978) ‘Information processing as an integrating concept in organizational design’, *Management Science*, 24(9), pp. 882–895. Available at: <https://pubsonline.informs.org/doi/abs/10.1287/mnsc.24.9.887> (Accessed: 18 April 2026).
- Gibbs, G. (1988) *Learning by Doing: A Guide to Teaching and Learning Methods*. Oxford: Oxford Polytechnic. Reprint available at: <https://dsdweb.co.uk/wp-content/uploads/2021/10/learning-by-doing-graham-gibbs.pdf> (Accessed: 26 April 2026).